

Credit Risk Classification Project Report

1. Introduction

Credit risk assessment plays a critical role in the banking and financial services industry. Financial institutions must evaluate the likelihood of a borrower defaulting on a loan before approval. Inaccurate risk assessment can lead to severe financial losses or missed revenue opportunities. This project aims to design an end-to-end machine learning-based credit risk classification system using the German Credit dataset, focusing on accuracy, recall of defaulters, interpretability, and deployability.

2. Dataset Description

The German Credit dataset consists of customer-level demographic and financial attributes. Key variables include age, sex, job type, housing status, saving accounts, checking account balance, credit amount, loan duration, and loan purpose. The target variable, Risk, indicates whether a customer represents a good credit risk (0) or a bad credit risk (1). The dataset is moderately imbalanced, reflecting real-world lending scenarios.

3. Exploratory Data Analysis (EDA)

Exploratory Data Analysis was conducted to gain insights into the dataset and uncover patterns. The analysis included examining the distribution of the target variable, visualizing numerical features using histograms, and identifying outliers through boxplots grouped by credit risk category. Correlation analysis was also performed to understand relationships between numerical variables. These insights guided feature engineering and model selection decisions.

4. Data Preprocessing

Data preprocessing was a crucial step to ensure model reliability. Missing values in Saving accounts and Checking account were imputed using the mode, as these features are categorical in nature. Categorical variables were transformed using one-hot encoding, while numerical features were standardized for models sensitive to feature scaling. The target variable was encoded as 0 for good credit and 1 for bad credit, aligning with binary classification requirements.

5. Feature Engineering

Feature engineering was performed to incorporate domain knowledge and enhance predictive power:

- `Credit_to_Duration_Ratio` captures the repayment burden relative to loan duration.
- `Age_Group` categorizes customers into meaningful age buckets to model non-linear age effects.
- `High_Risk_Purpose` flags loan purposes historically associated with higher default rates.
- `Account_Stability` combines saving and checking account information into a single stability

score.

These engineered features improved model interpretability and performance.

6. Model Development

Multiple classification algorithms were implemented to compare performance. Logistic Regression was used as a baseline model due to its simplicity and interpretability. Decision Tree provided a non-linear and interpretable approach. Random Forest and XGBoost, both ensemble-based models, were included to capture complex relationships and improve predictive accuracy. Pipelines were used to prevent data leakage and maintain consistency.

7. Hyperparameter Tuning

Hyperparameter tuning was conducted using GridSearchCV with 5-fold cross-validation for Random Forest and XGBoost models. Parameters such as number of estimators, tree depth, and minimum samples per split were optimized. Tuned models demonstrated improved generalization performance compared to their baseline counterparts.

8. Model Evaluation & Visualization

Model evaluation was carried out using multiple metrics: Accuracy, Precision, Recall, F1-score, and ROC-AUC. Confusion matrices were analyzed to understand classification errors. ROC curves and Precision-Recall curves provided threshold-based performance insights. Feature importance plots for tree-based models highlighted the most influential variables driving credit risk predictions.

9. Business Interpretation

From a business perspective, Recall for the default class is the most critical metric. False Negatives, where risky customers are incorrectly classified as safe, lead to direct financial losses and increased credit exposure. False Positives result in rejecting potentially good customers, which represents an opportunity cost but is less severe. A recall threshold between 0.75 and 0.85 is recommended to balance risk control and business growth.

10. Model Selection & Deployment Considerations

The tuned XGBoost model is recommended for production deployment. It achieved the best balance between performance and stability, while offering feature importance for interpretability. The model supports fast inference suitable for real-time credit scoring. Deployment can be achieved via a REST API using FastAPI, containerized with Docker, and hosted on cloud infrastructure. Continuous monitoring, data drift detection, scheduled retraining, and A/B testing are essential to ensure long-term reliability.

11. Conclusion

This project demonstrates a complete, production-oriented credit risk modeling pipeline. By combining thorough data analysis, robust feature engineering, advanced machine learning models, and business-driven evaluation criteria, the system provides a reliable solution for credit risk prediction. The tuned XGBoost model offers the most effective trade-off between risk mitigation and operational efficiency.